QMS Journal Club

Choong Pak Shen

March 22, 2026

QMS Journal Club 1st meeting

Linear algebra

A vector $\vec{v}$ is an n-tuple of entries, which can be represented by a column vector,

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \tag{1}$$

For our purposes, we focus only on complex vector space, which is the space of all complex vectors. The n-dimensional complex vector space is written as $\mathbb{C}^n$.

Vector space has two operations,

1. Vector addition, $\vec{v} = \vec{v}_1 + \vec{v}_2$
2. Scalar multiplication, $c\vec{v}$

A quantum bit (qubit) is a superposition (linear combination) of the basis states ($|0\rangle$, $|1\rangle$),

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{2}$$

where ψ_0 and ψ_1 are called probability amplitudes, which are generally complex. The notation $|\psi\rangle$ is called a ket vector and can be represented as a column vector,

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \tag{3}$$

where $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. $|0\rangle$ and $|1\rangle$ are orthonormal to each other.

The dual to the ket vector is a bra vector,

$$\langle\psi| = \bar{\psi}_0\langle 0| + \bar{\psi}_1\langle 1| = \begin{pmatrix} \bar{\psi}_0 & \bar{\psi}_1 \end{pmatrix}, \tag{4}$$

where the overhead bar is complex conjugate operation.

The state vectors of a qubit live in the Hilbert space, which is a two-dimensional complex vector space $\mathcal{H} = \mathbb{C}^2$. The Hilbert space is equipped with an inner product operation,

$$\begin{aligned} \langle\psi|\psi\rangle &= \begin{pmatrix}\bar{\psi}_0 & \bar{\psi}_1\end{pmatrix}\begin{pmatrix}\psi_0 \\ \psi_1\end{pmatrix} \\ &= |\psi_0|^2 + |\psi_1|^2. \end{aligned} \tag{5}$$

There is also an outer product (tensor product),

$$\begin{aligned} |\psi\rangle\langle\psi| &= \begin{pmatrix}\psi_0 \\ \psi_1\end{pmatrix}\begin{pmatrix}\bar{\psi}_0 & \bar{\psi}_1\end{pmatrix} \\ &= \begin{pmatrix}|\psi_0|^2 & \psi_0\bar{\psi}_1 \\ \psi_1\bar{\psi}_0 & |\psi_1|^2\end{pmatrix} \end{aligned} \tag{6}$$

There exists four extremely important matrices, called the Pauli matrices:

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{7}$$

$$\sigma_1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{8}$$

$$\sigma_2 = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{9}$$

$$\sigma_3 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10}$$

It is important to take note that

$$XY = iZ; YX = -iZ, \tag{11}$$

$$YZ = iX; ZY = -iX, \tag{12}$$

$$ZX = iY; XZ = -iY, \tag{13}$$

$$XX = YY = ZZ = I. \tag{14}$$

The Pauli matrices are traceless, i.e. the sum of diagonal elements is zero, $Tr(\sigma_i) = 0$.

It is also Hermitian, i.e. its conjugate transpose is equivalent to itself, $\sigma_i^\dagger = \sigma_i$, where $\dagger$ denotes conjugate transpose operation.

Let

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{15}$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}; |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \tag{16}$$

$$|+i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}; |-i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{17}$$

The Pauli matrices have two eigenvalues ± 1,

$$X|+\rangle = |+\rangle; X|-\rangle = -|-\rangle, \tag{18}$$
$$Y|+i\rangle = |+i\rangle; Y|-i\rangle = -|-i\rangle, \tag{19}$$
$$Z|0\rangle = |0\rangle; Z|1\rangle = -|1\rangle. \tag{20}$$

The Pauli matrices can be formulated by outer products,

$$X = |+\rangle\langle+| - |-\rangle\langle-|, \tag{21}$$
$$Y = |+i\rangle\langle+i| - |-i\rangle\langle-i|, \tag{22}$$
$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|. \tag{23}$$

Previously, we mentioned that $\langle\psi|$ is the dual to $|\psi\rangle$. By using conjugate transpose operation, they are related by

$$\langle\psi| = (|\psi\rangle)^\dagger. \tag{24}$$

If A and B are two matrices, then $(AB)^\dagger = B^\dagger A^\dagger$. Therefore, $(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger$.

Also, $(A+B)^\dagger = A^\dagger + B^\dagger$. Using this fact, one can easily show that the Pauli matrices are Hermitian, $\sigma_i^\dagger = \sigma_i$.

A matrix is unitary if $U^\dagger = U^{-1}$. The Pauli matrices are Hermitian and unitary.

Tensor product manifests itself in matrices as Kronecker product. Given two matrices A and B,

$$A \otimes B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \otimes \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \tag{25}$$

In general, $A \otimes B \neq B \otimes A$.

If we have three vectors $|v_1\rangle$, $|v_2\rangle \in V$ and $|w\rangle \in W$,

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle = |v_1 w\rangle + |v_2 w\rangle, \tag{26}$$

$$|w\rangle \otimes (|v_1\rangle + |v_2\rangle) = |w\rangle \otimes |v_1\rangle + |w\rangle \otimes |v_2\rangle = |w v_1\rangle + |w v_2\rangle. \tag{27}$$

If A acts on the vector space V and B acts on the vector space W, then

$$(A \otimes B)|v_1 w\rangle = A|v_1\rangle \otimes B|w\rangle. \tag{28}$$

Question: The Hadamard operator on one-qubit can be written as

$$H = \frac{1}{\sqrt{2}}\left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|\right]. \tag{29}$$

Show that the Hadamard transform on two qubits can be written as

$$H^{\otimes 2} = H \otimes H = \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0,1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle\langle y_1 y_2|. \tag{30}$$

$$
\begin{aligned}
H \otimes H &= \frac{1}{\sqrt{2}}\left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|\right] \otimes \\
&\quad \frac{1}{\sqrt{2}}\left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|\right] \\
&= \frac{1}{2}[(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\langle 00| \\
&\quad + (|00\rangle - |01\rangle + |10\rangle - |11\rangle)\langle 01| \\
&\quad + (|00\rangle + |01\rangle - |10\rangle - |11\rangle)\langle 10| \\
&\quad + (|00\rangle - |01\rangle - |10\rangle + |11\rangle)\langle 11|] \\
&= \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0,1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle\langle y_1 y_2|.
\end{aligned}
$$

The trace of a matrix A is defined as the sum of its diagonal elements,

$$
\begin{aligned}
Tr(A) = Tr(IAI) &= Tr\left(\sum_{jk} |j\rangle\langle j|A|k\rangle\langle k|\right) \\
&= Tr\left(\sum_{jk} A_{jk}|j\rangle\langle k|\right) = \sum_{ijk} A_{jk}\langle i|j\rangle\langle k|i\rangle = \sum_{ijk} A_{jk}\langle k|i\rangle\langle i|j\rangle \\
&= \sum_{jk} A_{jk}\langle k|j\rangle = \sum_{jk} A_{jk}\delta_{kj} = \sum_{j} A_{jj}. \qquad (31)
\end{aligned}
$$

The trace of a matrix satisfies a few properties:

1. $Tr(AB) = Tr(BA)$
2. $Tr(ABC) = Tr(BCA) = Tr(CAB)$
3. $Tr(A + B) = Tr(A) + Tr(B)$
4. $Tr(cA) = cTr(A)$
5. $Tr(A^T) = Tr(A)$
6. $Tr(A^T B) = Tr(AB^T) = Tr(B^T A) = Tr(BA^T)$
7. $Tr(A \otimes B) = Tr(A)Tr(B)$

Suppose $|\psi\rangle$ is a unit vector and A is an arbitrary operator.

$$\begin{aligned} Tr(A|\psi\rangle\langle\psi|) &= \sum_i \langle i|A|\psi\rangle\langle\psi|i\rangle \\ &= \sum_i \langle\psi|i\rangle\langle i|A|\psi\rangle \\ &= \langle\psi|A|\psi\rangle. \end{aligned} \tag{32}$$

QMS Journal Club 2nd meeting

Postulates of quantum mechanics

Postulate 1: Associated to any isolated physical system is a complex vector space with inner product, i.e. a Hilbert space, known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the systems' state space.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2,

$$|\psi'\rangle = U|\psi\rangle. \tag{33}$$

Explicitly, it can be described by the Schrodinger equation,

$$i\hbar\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle. \tag{34}$$

Or,

$$|\psi(t_2)\rangle = e^{\frac{-i\hat{H}(t_2-t_1)}{\hbar}}|\psi(t_1)\rangle. \tag{35}$$

Question: Suppose A and B are commuting Hermitian operators. Show that $e^A e^B = e^{(A+B)}$ by expanding the left and right hand sides until $n = 3$.

Question: Show that e^{iK} is unitary for some Hermitian matrix K. You only need to expand until $n = 6$.

Note that the matrix exponential is given by

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \tag{36}$$

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle, \tag{37}$$

and the state of the system after the measurement is

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}. \tag{38}$$

The measurement operators satisfies the completeness equation,

$$\sum_m M_m^\dagger M_m = I. \tag{39}$$

A projective measurement is described by an observable, M, a Hermitian operator on the state space of the system being observed. The observable can be written as

$$M = \sum_m m|m\rangle\langle m|, \tag{40}$$

where $P_m = |m\rangle\langle m|$ is the projector onto the eigenspace of M with eigenvalue m, satisfying $P_m P_{m'}^\dagger = \delta_{mm'} P_m$.
The expected value of the measurement is

$$\begin{aligned} E(M) = \langle M\rangle &= \sum_m m p(m) \\ &= \sum_m m\langle\psi|P_m|\psi\rangle \\ &= \langle\psi|\sum_m m|m\rangle\langle m||\psi\rangle \\ &= \langle\psi|M|\psi\rangle \end{aligned} \tag{41}$$

Question: The variance associated to observations of M, $[\Delta(M)]^2$ is given by

$$[\Delta(M)]^2 = \left\langle (M - \langle M \rangle)^2 \right\rangle . \tag{42}$$

Show that it is equivalent to $[\Delta(M)]^2 = \left\langle M^2 \right\rangle - \langle M \rangle^2$.

Question: Suppose A and B are two Hermitian operators, and $\langle\psi|AB|\psi\rangle = x + iy$. Show that $\langle\psi|[A,B]|\psi\rangle = 2iy$, $\langle\psi|\{A,B\}|\psi\rangle = 2x$.

The above implies that

$$|\langle\psi|[A,B]|\psi\rangle|^2 + |\langle\psi|\{A,B\}|\psi\rangle|^2 = 4|\langle\psi|AB|\psi\rangle|^2. \tag{43}$$

By the Cauchy-Schwarz inequality,

$$|\langle\psi|AB|\psi\rangle|^2 \leq \langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle. \tag{44}$$

Hence,

$$|\langle\psi|[A,B]|\psi\rangle|^2 \leq 4\langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle. \tag{45}$$

Question: Suppose C and D are two observables, and $A = C - \langle C \rangle$, $B = D - \langle D \rangle$. Obtain the Heisenberg's uncertainty principle,

$$\Delta(C)\Delta(D) \geq \frac{|\langle \psi | [C, D] | \psi \rangle|}{2}. \tag{46}$$

QMS Journal Club 3rd meeting

Phase

Global phase factor is a term $e^{i\theta}$ attached to $|\psi\rangle$ without changing the measurement statistics of $|\psi\rangle$,

$$\langle\psi|e^{-i\theta}M_m^\dagger M_m e^{i\theta}|\psi\rangle = \langle\psi|M_m^\dagger M_m|\psi\rangle. \tag{47}$$

For this reason, we say that $|\psi\rangle \sim e^{i\theta}|\psi\rangle$.

Relative phase is a term $e^{i\theta}$ attached between probability amplitudes,

$$|\psi\rangle = \alpha|\psi_\alpha\rangle + e^{i\theta}\beta|\psi_\beta\rangle. \tag{48}$$

Relative phase is a basis-dependent concept and may give rise to physically observable differences in measurement statistics.

For example, $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ are not the same up to a relative phase shift.

For one quantum bit (qubit), the Hilbert space is given as $\mathcal{H} = \mathbb{C}^2$, where

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{49}$$

$$= \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \tag{50}$$

where ψ_0 and ψ_1 are complex numbers, and $|\psi_0|^2 + |\psi_1|^2 = 1$.

At first glance, there are 4 degrees of freedom. However, due to the normalization condition and the equivalence of global phase factor, we only need 2 degrees of freedom to parametrize a one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{51}$$

Given the following Pauli matrices, find the expected values of Pauli-X, Y, and Z for $|\psi\rangle$.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \; Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \; Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{52}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{53}$$

$$\langle\psi|X|\psi\rangle = \tag{54}$$

$$\langle\psi|Y|\psi\rangle = \tag{55}$$

$$\langle\psi|Z|\psi\rangle = \tag{56}$$

The expected values of Pauli matrices correspond to the spherical coordinates when the radius equals to 1,

$$\langle\psi|X|\psi\rangle = \sin\theta\cos\phi, \quad (57)$$

$$\langle\psi|Y|\psi\rangle = \sin\theta\sin\phi, \quad (58)$$

$$\langle\psi|Z|\psi\rangle = \cos\theta. \quad (59)$$

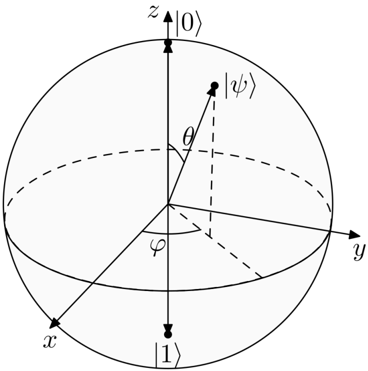


Figure 1: Bloch sphere

Postulates of quantum mechanics

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \ldots \otimes |\psi_n\rangle. \tag{60}$$

Postulate 4 enables us to define quantum entanglement. Suppose we are given two qubit states, $|\psi\rangle \in \mathcal{H}_A = \mathbb{C}^2$ and $|\phi\rangle \in \mathcal{H}_B = \mathbb{C}^2$. The two-qubit state $|\psi_{AB}\rangle$ is entangled if $|\psi_{AB}\rangle \neq |\psi\rangle \otimes |\phi\rangle$.
For example, the following state $|\psi_{AB}\rangle$ is separable,

$$\begin{aligned}
|\psi_{AB}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\
&= \frac{1}{2}[|0\rangle \otimes (|0\rangle + |1\rangle) + |1\rangle \otimes (|0\rangle + |1\rangle)] \\
&= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\
&= |+\rangle \otimes |+\rangle
\end{aligned}$$

It is not always obvious to identify if a given two-qubit state is separable or not. One can derive separability condition for two qubits. As long as the two-qubit state satisfies the separability condition, the two-qubit state is separable (else, it is entangled).

Let $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$, $|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle$. Then

$$\begin{aligned}
|\psi_{AB}\rangle &= |\psi\rangle \otimes |\phi\rangle \\
&= (\psi_0|0\rangle + \psi_1|1\rangle) \otimes (\phi_0|0\rangle + \phi_1|1\rangle) \\
&= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle
\end{aligned}$$

Notice that the probability amplitudes of $|00\rangle$ and $|11\rangle$, when multiplied together, is the same as the product of probability amplitudes of $|10\rangle$ and $|01\rangle$. We can conclude that for a two-qubit state $|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$, if $\psi_{00}\psi_{11} = \psi_{10}\psi_{01}$, the two-qubit state is separable. Else, it is entangled.

Superdense coding

Can Alice communicate a number of classical bits of information by transmitting a smaller number of qubits, under the assumption that Alice and Bob receive an entangled resource beforehand?

- Alice and Bob initially share a pair of qubits in the Bell state,

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}; \tag{61}$$

- If Alice wants to send 00, she does nothing; If Alice wants to send 01, she applies Z; If Alice wants to send 10, she applies X; If Alice wants to send 11, she applies iY;

- The final two-qubit state becomes:

$$00 : |\psi\rangle \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{62}$$

$$01 : |\psi\rangle \rightarrow \frac{|00\rangle - |11\rangle}{\sqrt{2}} \tag{63}$$

$$10 : |\psi\rangle \rightarrow \frac{|10\rangle + |01\rangle}{\sqrt{2}} \tag{64}$$

$$11 : |\psi\rangle \rightarrow \frac{|01\rangle - |10\rangle}{\sqrt{2}} \tag{65}$$

- Alice sends her qubit to Bob;
- Bob performs a Bell basis measurement to determine which of the four possible bit strings Alice sent.

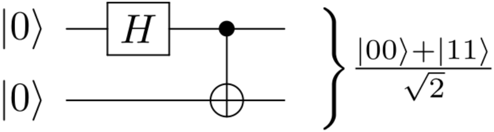


Figure 2: Bell state creation

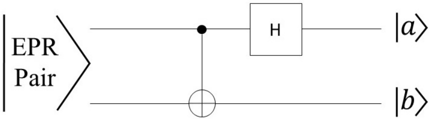


Figure 3: Bell basis measurement

QMS Journal Club 4th meeting

Recap

1. A brief review on linear algebra
    - Basic operations on vectors and matrices
    - Pauli matrices
    - Inner product and outer product
    - Tensor product and Kronecker product
    - Trace of a matrix
2. Postulates of quantum mechanics
    - State space of a closed quantum system
    - Dynamics of a closed quantum system
    - Quantum measurement
    - Composite quantum system
    - Superdense coding

Consider a vertically polarized photon, given by

$$|V\rangle = \frac{|L\rangle + |R\rangle}{\sqrt{2}}. \tag{66}$$

When the photon passes through a circular polarizer that allows either only $|L\rangle$ or only $|R\rangle$ polarized light, half of the photons will be absorbed in both cases.

This may seem like half of the photons are in $|L\rangle$ or $|R\rangle$, but this is not true.

If the photon passes through a linear polarizer, there will be no absorption.

Meanwhile, if we pass unpolarized light through any type of polarizer, half of the intensity of light will be lost.

For the case of polarized light, the action of “measuring” the polarization of light is contextual, i.e. the outcome depends on what kind of measurement is being performed.

For the case of unpolarized light, the action of “measuring” the polarization of light is not contextual, because the outcome does not depend on the type of measurement being performed.

This suggests that for the case of polarized light, a quantum treatment is required (contextuality is a quantum phenomenon), whereas a classical treatment is required for the case of unpolarized light.

Polarized light is one example of pure states, whereby the probability of obtaining $|L\rangle$ or $|R\rangle$ is given by the absolute square of the probability amplitude.

Unpolarized light is one example of mixed states, and it is considered as a (classical) statistical ensemble, i.e. each photon having either $|R\rangle$ or $|L\rangle$ polarization with probability $\frac{1}{2}$.

For this example,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{67}$$

These two ensembles are completely indistinguishable experimentally, hence they are considered the same mixed state.

Conventionally,

$$|H\rangle = |0\rangle,\, |V\rangle = |1\rangle, \tag{68}$$

$$|D\rangle = |+\rangle,\, |A\rangle = |-\rangle, \tag{69}$$

$$|L\rangle = |+i\rangle,\, |R\rangle = |-i\rangle. \tag{70}$$

Formally, the density matrix of a quantum system is defined as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \tag{71}$$

where $\sum_i p_i = 1$. We say that ρ is a mixture of pure states.

Density matrix is a more general formalism compared to state vector formalism, since a pure state $|\psi\rangle$ can be described by a density matrix $\rho = |\psi\rangle\langle\psi|$.

Consider

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{72}$$
$$|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle, \tag{73}$$
$$\rho = p_\psi|\psi\rangle\langle\psi| + p_\phi|\phi\rangle\langle\phi|, \tag{74}$$

where $p_\psi + p_\phi = 1$. Find ρ in a 2×2 matrix.

The density matrix is given by

$$\rho = \begin{pmatrix} p_\psi|\psi_0|^2 + p_\phi|\phi_0|^2 & p_\psi\psi_0\bar{\psi}_1 + p_\phi\phi_0\bar{\phi}_1 \\ p_\psi\bar{\psi}_0\psi_1 + p_\phi\bar{\phi}_0\phi_1 & p_\psi|\psi_1|^2 + p_\phi|\phi_1|^2 \end{pmatrix}. \tag{75}$$

If we compute $\langle 0|\rho|0\rangle$, we get $p_\psi|\psi_0|^2 + p_\phi|\phi_0|^2$.

The diagonal element gives the probability of finding a particle in the corresponding basis state $\{|0\rangle, |1\rangle\}$.

For a general state $|\Psi\rangle$,

$$\begin{aligned} \langle\Psi|\rho|\Psi\rangle &= p_\psi\langle\Psi|\psi\rangle\langle\psi|\Psi\rangle + p_\phi\langle\Psi|\phi\rangle\langle\phi|\Psi\rangle \\ &= p_\psi|\langle\psi|\Psi\rangle|^2 + p_\phi|\langle\phi|\Psi\rangle|^2 \end{aligned} \tag{76}$$

it can be seen that $\langle\Psi|\rho|\Psi\rangle$ gives the total probability of finding a particle in the pure state $|\Psi\rangle$ within a mixture ρ.

A density matrix ρ has the following properties:

1. $Tr(\rho) = 1$;
2. $\rho^\dagger = \rho$;
3. It is a positive semi-definite matrix, i.e. $\langle\psi|\rho|\psi\rangle \geq 0$.

For pure states, the density matrix ρ further satisfies the following:

1. $\rho^2 = \rho$;
2. $Tr(\rho^2) = 1$.

How many real parameters do you need to describe a 2×2 density matrix?

You need three real parameters to describe a 2×2 density matrix.

Consider again the one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{77}$$

Its density matrix is given by

$$\rho = \begin{pmatrix} \cos^2\left(\frac{\theta}{2}\right) & e^{-i\phi}\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) & \sin^2\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{78}$$

Previously, we learn that

$$P_x = \langle\psi|X|\psi\rangle = \sin\theta\cos\phi, \tag{79}$$

$$P_y = \langle\psi|Y|\psi\rangle = \sin\theta\sin\phi, \tag{80}$$

$$P_z = \langle\psi|Z|\psi\rangle = \cos\theta. \tag{81}$$

We want to rewrite the density matrix in terms of $\{P_x, P_y, P_z\}$, which are called the polarization (Bloch) vectors,

$$\begin{aligned}\rho &= \begin{pmatrix} \frac{\cos\theta+1}{2} & \frac{\sin\theta(\cos\phi - i\sin\phi)}{2} \\ \frac{\sin\theta(\cos\phi+i\sin\phi)}{2} & \frac{1-\cos\theta}{2} \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} P_z+1 & P_x - iP_y \\ P_x + iP_y & 1-P_z \end{pmatrix} \\ &= \frac{1}{2}\left(I + P_x X + P_y Y + P_z Z\right) \\ &= \frac{1}{2}\left(I + \vec{P}\cdot\vec{\sigma}\right),\end{aligned} \tag{82}$$

where

$$\vec{P} = \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix}, \vec{\sigma} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \tag{83}$$

In general, a density matrix can be written as $\rho = \frac{1}{2}\left(I + \vec{r} \cdot \vec{\sigma}\right)$.

Density matrix formalism is useful when

1. the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
2. when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).

QMS Journal Club 5th meeting

Previously, we mentioned that density matrix formalism is useful when

1. the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
2. when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).

Consider the Bell state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{84}$$

The description of this Bell state is that when measurement is performed for qubit A, there is a probability $\frac{1}{2}$ to get $|0\rangle$ or $|1\rangle$.

How can we describe the quantum state for qubit A alone?

One might be tempted to write

$$|\psi_A\rangle = \frac{1}{\sqrt{2}}(|0_A\rangle + |1_A\rangle). \tag{85}$$

This description gives you the correct statistics when you measure in the Pauli Z basis, but it gives you a wrong statistics when you measure in the Pauli X basis because $|\psi_A\rangle$ is now its eigenstate, $|+\rangle$.

This again shows that the state vector formalism does not completely capture the classical statistical ensemble aspect of mixed states.

Previously, an example of maximally mixed state was given,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{86}$$

Regardless of what measurement basis we are taking, we always get a probability of $\frac{1}{2}$ to get either $\{|R\rangle, |L\rangle\}$ (or $\{|H\rangle, |V\rangle\}$).

Postulates of quantum mechanics

Postulate 1: Associate to any isolated physical system is a complex vector space with inner product, known as the state space of the system. The system is completely described by its density operator, which is a positive semi-definite operator $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with trace one, acting on the state space of the system.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. The state ρ of the system at time t_1 is related to the state ρ' of the system at time t_2 by a unitary operator U,

$$\rho' = U \rho U^\dagger. \tag{87}$$

Postulate 3: Quantum measurements are described by a collection of measurement operators M_m, such that the probability that result m occurs for the state of a quantum system ρ is given by

$$p(m) = Tr(M_m^\dagger M_m \rho), \tag{88}$$

and the state of the system after the measurement is

$$\frac{M_m \rho M_m^\dagger}{Tr(M_m^\dagger M_m \rho)}. \tag{89}$$

For projective measurement $M = \sum_m m|m\rangle\langle m|$ with projector $P_m = |m\rangle\langle m|$, $P_m^\dagger P_m = P_m$ and hence

$$\begin{aligned} p(m) &= \sum_i p_i \langle\psi_i|P_m|\psi_i\rangle = \sum_i p_i \langle\psi_i|P_m \sum_j |j\rangle\langle j|\psi_i\rangle \\ &= \sum_{ij} p_i \langle j|\psi_i\rangle\langle\psi_i|P_m|j\rangle = Tr\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|P_m\right) \\ &= Tr(\rho P_m) = Tr(P_m \rho). \end{aligned} \tag{90}$$

The expectation value of a measurement M follows the same argument and can be written as

$$\langle M \rangle = Tr(M\rho). \tag{91}$$

Postulate 4: The state space of a composite quantum system is the tensor product of the state spaces of the component quantum systems, such that the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \ldots \otimes \rho_n$.
Suppose that a density matrix is given as

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|. \tag{92}$$

Let

$$|a\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle, \tag{93}$$

$$|b\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle. \tag{94}$$

Then,

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1| = \frac{1}{2}|a\rangle\langle a| + \frac{1}{2}|b\rangle\langle b|. \tag{95}$$

Different ensembles of quantum states can give rise to the same density matrix. When will two sets of state vectors $|\psi\rangle$ and $|\phi\rangle$ generate the same density matrix?

Theorem: The sets $|\psi_i\rangle$ and $|\phi_j\rangle$ generate the same density matrix if and only if

$$|\psi_i\rangle = \sum_j U_{ij}|\phi_j\rangle, \tag{96}$$

where U_{ij} is a unitary matrix.

Reduced density matrix

Suppose we have a quantum system A and B, defined by ρ_{AB}. The reduced density operator for system A is defined by

$$\rho_A = Tr_B(\rho_{AB}), \tag{97}$$

where Tr_B is called partial trace over system B. Partial trace is defined by

$$Tr_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| Tr(|b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2|\langle b_2|b_1\rangle. \tag{98}$$

Question 1: Consider the Bell state $|\psi_{AB}\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$. Find the reduced density matrix of the second qubit.

Question 2: Suppose a composite quantum system A and B is in the state $|a\rangle|b\rangle$. Show that the reduced density matrix of system A is a pure state.

Quantum teleportation

Quantum teleportation begins with two parties, Alice and Bob, sharing a pair of entangled qubits in one of the Bell bases,

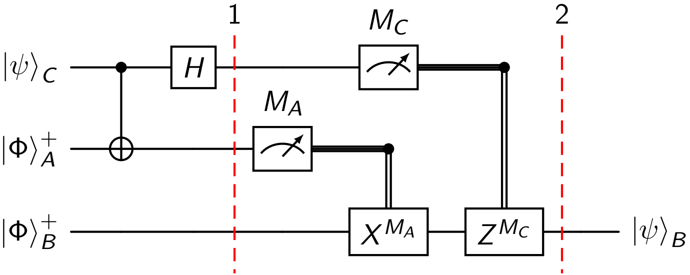
$$|\Phi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{99}$$

$$|\Phi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{100}$$

$$|\Psi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{101}$$

$$|\Psi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{102}$$

Without loss of generality, we focus on the Bell basis $|\Phi\rangle_{AB}^{+}$ in our discussion. The other Bell bases follow the same mathematical arguments.



1
2
M_C
M_A
$|\psi\rangle_C$
H
$|\Phi\rangle_A^+$
$|\Phi\rangle_B^+$
X^{M_A}
Z^{M_C}
$|\psi\rangle_B$

Initially, the shared Bell state between Alice and Bob, and the qubit that Alice wants to send to Bob is given as:

$$\begin{aligned} |\psi\rangle_{CAB} &= (\alpha\,|0\rangle_C + \beta\,|1\rangle_C) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \\ &= \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|100\rangle + \beta\,|111\rangle). \end{aligned} \tag{103}$$

After the CNOT gate, the above quantum state becomes:

$$|\psi\rangle_{CAB} = \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|110\rangle + \beta\,|101\rangle). \tag{104}$$

After the Hadamard gate, the quantum state at Step 1 will become:

$$
\begin{aligned}
|\psi\rangle_{CAB} = \frac{1}{\sqrt{2}} \Bigg[& \frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |00\rangle_{AB} + \frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |11\rangle_{AB} \\
& + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |10\rangle_{AB} + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |01\rangle_{AB} \Bigg] \\
= \frac{1}{2}[\alpha\,|000\rangle & + \alpha\,|100\rangle + \alpha\,|011\rangle + \alpha\,|111\rangle + \beta\,|010\rangle - \beta\,|110\rangle \\
& + \beta\,|001\rangle - \beta\,|101\rangle] \\
= \frac{1}{2}[|00\rangle_{CA} & (\alpha\,|0\rangle_B + \beta\,|1\rangle_B) + |01\rangle_{CA}(\beta\,|0\rangle_B + \alpha\,|1\rangle_B) \\
& + |10\rangle_{CA}(\alpha\,|0\rangle_B - \beta\,|1\rangle_B) + |11\rangle_{CA}(-\beta\,|0\rangle_B + \alpha\,|1\rangle_B)].
\end{aligned}
\tag{105}
$$

In Step 2, quantum measurements will be performed on Alice's qubits (qubit A and C) and the appropriate Pauli correction will be applied to Bob's qubit (qubit B) to recover the quantum information that is teleported. In short,

1. If the quantum measurement results in $|00\rangle_{CA}$, nothing ($\sigma_0 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$) will be performed;
2. If the quantum measurement results in $|01\rangle_{CA}$, Pauli X, $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ will be performed;
3. If the quantum measurement results in $|10\rangle_{CA}$, Pauli Z, $\sigma_3 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ will be performed;
4. If the quantum measurement results in $|11\rangle_{CA}$, Pauli Y up to a global phase of i, $i\sigma_2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ will be performed.

Find the density matrix of $|\psi\rangle_{CAB}$,

$$|\psi\rangle_{CAB} = \frac{1}{2}[|00\rangle\,(\alpha\,|0\rangle + \beta\,|1\rangle) + |01\rangle\,(\beta\,|0\rangle + \alpha\,|1\rangle)$$
$$+\,|10\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) + |11\rangle\,(-\beta\,|0\rangle + \alpha\,|1\rangle)]. \tag{106}$$

Verify that when Alice's system is being traced out, the reduced density matrix of Bob is maximally mixed.

This shows that any measurements performed by Bob will contain no information about $|\psi\rangle_{CAB}$, and Alice cannot use quantum teleportation to transmit information to Bob faster than speed of light.

Here are some intermediate steps for your verifications. Let

$$
\begin{aligned}
|B_1\rangle &= \alpha\,|0\rangle + \beta\,|1\rangle\,, \\
|B_2\rangle &= \beta\,|0\rangle + \alpha\,|1\rangle\,, \\
|B_3\rangle &= \alpha\,|0\rangle - \beta\,|1\rangle\,, \\
|B_4\rangle &= -\beta\,|0\rangle + \alpha\,|1\rangle\,.
\end{aligned}
$$

Then

$$
\begin{aligned}
\rho_{AB} = \frac{1}{4}[&|0\rangle\langle 0| \otimes |B_1\rangle\langle B_1| + |0\rangle\langle 1| \otimes |B_1\rangle\langle B_2| + |1\rangle\langle 0| \otimes |B_2\rangle\langle B_1| \\
&+ |1\rangle\langle 1| \otimes |B_2\rangle\langle B_2| + |0\rangle\langle 0| \otimes |B_3\rangle\langle B_3| + |0\rangle\langle 1| \otimes |B_3\rangle\langle B_4| \\
&+ |1\rangle\langle 0| \otimes |B_4\rangle\langle B_3| + |1\rangle\langle 1| \otimes |B_4\rangle\langle B_4|]. \qquad (107)
\end{aligned}
$$

Lastly,

$$
\begin{aligned}
\rho_B &= \frac{1}{4}[|B_1\rangle\langle B_1| + |B_2\rangle\langle B_2| + |B_3\rangle\langle B_3| + |B_4\rangle\langle B_4|] \\
&= \frac{1}{4}[2(|\alpha|^2 + |\beta|^2)|0\rangle\langle 0| + 2(|\alpha|^2 + |\beta|^2)|1\rangle\langle 1|] = \frac{1}{2}I. \qquad (108)
\end{aligned}
$$

QMS Journal Club 6th meeting

Higher order tensors

From one-qubit state

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle \tag{109}$$

to two-qubit state

$$|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle, \tag{110}$$

the probability amplitude of the quantum states changes from an one-index object to a two-index object.
In general, one-index object $\Psi_A = (\psi_i)$ is called tensor of order 1, while two-index object $\Psi_{AB} = (\psi_{ij})$ is called tensor of order 2.
Tensor of order 1 is usually represented as a column vector and it coincides with the state vector representation.
Meanwhile, tensor of order 2 is represented as a matrix. In two-qubit case,

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{111}$$

Schmidt decomposition

Theorem: Suppose $|\psi\rangle$ is a pure state of a composite system AB. Then there exist orthonormal states $|i_A\rangle$ for system A, and orthonormal states $|i_B\rangle$ for system B, such that

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle|i_B\rangle, \tag{112}$$

where λ_i are non-negative real numbers satisfying $\sum_i \lambda_i^2 = 1$, known as Schmidt coefficients.

Schmidt decomposition is a consequence of singular value decomposition when representing $|\psi\rangle$ as a tensor of order 2,

$$\Psi_{AB} = U\Lambda V^T, \tag{113}$$

where U and V are unitary matrices and Λ is a diagonal matrix.

Singular value decomposition works for rectangular matrices. As a consequence of this, Schmidt decomposition works for any bipartite quantum systems of different dimension, for example qubit-qutrit system.
As a convention, we usually require an ordering condition, i.e. $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$. This makes sure that the unitary matrices U and V to be uniquely determined.
For two qubits, the reduced density matrices ρ_A and ρ_B are

$$\rho_A = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\bar{\psi}_{10} + \psi_{01}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{10} + \bar{\psi}_{01}\psi_{11} & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}, \tag{114}$$

$$\rho_B = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{10}|^2 & \psi_{00}\bar{\psi}_{01} + \psi_{10}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{01} + \bar{\psi}_{10}\psi_{11} & |\psi_{01}|^2 + |\psi_{11}|^2 \end{pmatrix}. \tag{115}$$

Question: Show that

$$\rho_A = \Psi_{AB}\Psi_{AB}^\dagger, \tag{116}$$

$$\rho_B = \Psi_{AB}^T\bar{\Psi}_{AB}. \tag{117}$$

As a result, when singular value decomposition is applied,

$$\rho_A = \Psi_{AB}\Psi_{AB}^{\dagger} = U\Lambda V^T \bar{V}\Lambda U^{\dagger} = U\Lambda^2 U^{\dagger}, \tag{118}$$

$$\rho_B = \Psi_{AB}^T \bar{\Psi}_{AB} = V\Lambda U^T \bar{U}\Lambda V^{\dagger} = V\Lambda^2 V^{\dagger}. \tag{119}$$

This shows that U and V are the eigendecomposition of the reduced density matrices ρ_A and ρ_B respectively, and the reduced density matrices ρ_A and ρ_B have the same eigenvalues (singular values or Schmidt coefficients).

For two qubits, there are two Schmidt coefficients λ_1 and λ_2, and we know that $\lambda_1^2 + \lambda_2^2 = 1$. Therefore, there are only three possibilities,

1. Maximally entangled states: $\lambda_1^2 = \lambda_2^2 = \frac{1}{2}$
2. Entangled states: $\lambda_1^2 > \lambda_2^2 \neq 0$
3. Separable states: $\lambda_1^2 = 1$, $\lambda_2^2 = 0$

Question: Classify the following two-qubit states according to the framework.

1. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
2. $\frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$
3. $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

One can also describe the separability of two-qubit states by finding the separability condition. Let $|\psi_A\rangle \in \mathcal{H}_A$ and $|\phi_B\rangle \in \mathcal{H}_B$, and

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{120}$$

$$|\phi_B\rangle = \phi_0|0\rangle + \phi_1|1\rangle. \tag{121}$$

The combined two-qubit state becomes

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi_A\rangle \otimes |\phi_B\rangle \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle. \end{aligned} \tag{122}$$

One can notice that the product of probability amplitudes of $|00\rangle$ and $|11\rangle$ is the same as the product of probability amplitudes of $|01\rangle$ and $|10\rangle$, when the two-qubit state is separable.

Therefore, the separability condition of two qubits is given by

$$\psi_{00}\psi_{11} - \psi_{01}\psi_{10}. \tag{123}$$

If $\psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$, the two-qubit state is separable. Otherwise, it is entangled.

Last but not least, notice that the separability condition of two qubits is the same as the determinant of Ψ_{AB},

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{124}$$

The whole process is called entanglement classification of two qubits, which is well-studied for bipartite systems due to Schmidt decomposition.

From group theoretic perspective, the action of U and V is called local unitary group action on two qubits. Notice that U only acts on qubit A, V only acts on qubit B. Hence, singular value decomposition (or equivalently, Schmidt decomposition) on a two-qubit state can be written as

$$
\begin{aligned}
U \otimes V|\psi_{AB}\rangle &= \sum_{ij} U \otimes V \psi_{ij}|ij\rangle \\
&= \sum_{ijklmn} \psi_{ij}|k\rangle\langle k|U|m\rangle\langle m| \otimes |l\rangle\langle l|V|n\rangle\langle n|(|i\rangle \otimes |j\rangle) \\
&= \sum_{ijklmn} \psi_{ij}|k\rangle\langle k|U|m\rangle\langle m|i\rangle \otimes |l\rangle\langle l|V|n\rangle\langle n|j\rangle \\
&= \sum_{ijkl} \psi_{ij}|k\rangle\langle k|U|i\rangle \otimes |l\rangle\langle l|V|j\rangle \\
&= \sum_{ijkl} \psi_{ij} u_{ki} v_{lj}|k\rangle \otimes |l\rangle = \sum_{ijkl} u_{ki}\psi_{ij} v_{jl}^T |k\rangle \otimes |l\rangle
\end{aligned} \tag{125}
$$

Question: How many real parameters do you need to write down a 2×2 unitary matrix of determinant 1?

You need 3 real parameters to define a 2×2 unitary matrix,

$$U = \begin{pmatrix} u_{11} & u_{12} \\ -\bar{u}_{12} & \bar{u}_{11} \end{pmatrix}. \tag{126}$$

In order to specify a generic two-qubit state, you need 6 real parameters (ignoring the global phase).

Hence, it is possible for the local unitary group of two qubits to decompose a generic two-qubit state into the Schmidt coefficients, which are real and non-negative.

Question: Can we have Schmidt decomposition for three qubits?

State purification

Suppose we are given a mixed state ρ_A. It is possible to introduce another (reference) system, R, in such a way that the combined quantum state $|AR\rangle$ is a pure state, and $\rho_A = Tr_R(|AR\rangle\langle AR|)$.
Due to Schmidt decomposition, ρ_A can be written as

$$\rho_A = \sum_i p_i |i_A\rangle\langle i_A|. \tag{127}$$

The combined system needs to be in this form

$$|AR\rangle = \sum_i \sqrt{p_i}|i_A\rangle|i_R\rangle. \tag{128}$$

The partial trace Tr_R of the combined system is given by

$$\begin{aligned} Tr_R(|AR\rangle\langle AR|) &= \sum_{ij} \sqrt{p_i p_j}|i_A\rangle\langle j_A| Tr(|i_R\rangle\langle j_R|) = \sum_{ij} \sqrt{p_i p_j}|i_A\rangle\langle j_A|\delta_{ij} \\ &= \sum_i p_i |i_A\rangle\langle i_A| = \rho_A \end{aligned} \tag{129}$$

QMS Journal Club 7th meeting

Quantum contextuality

What is considered as “physical”?

Before quantum theory is developed, the intuition that we have is that the physical properties of an object exist independent of observation.

Measurements merely act to reveal such physical properties.

According to quantum theory, an unobserved particle does not possess physical properties that exist independent of observation.

Such physical properties arise as a consequence of measurements performed upon the system.

In order to counter this argument, one proposed the non-contextual hidden-variable theory, where

1. All quantum observables may be simultaneously assigned definite values, which may deterministically depend on some “hidden” classical variable;
2. Value assignments pre-exist and are independent of the choice of any other observables;
3. Some functional constraints on the assignments of values for compatible observables are assumed.

Kochen and Specker proved that such hidden variable theory does not exist, i.e. it is impossible to assign definite values to an observable independent of other compatible observables (contexts).

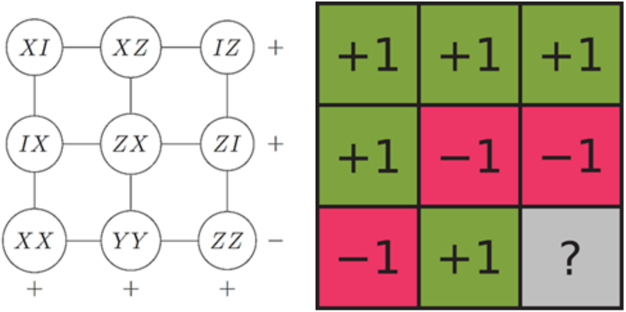


Figure 4: Mermin-Peres square.

Each line represents one context (mutually commuting operators) and each operator squares to identity, meaning that its eigenvalues are ± 1.

The product of the operators on each line is $I_1 I_2$, except for the last vertical context.

There is no way to assign a definite value ± 1 to each operator to reproduce these product rules because each operator appears in exactly two contexts.

EPR paradox

In a complete physical theory, there is an element corresponding to each element of reality. A sufficient condition for the reality of a physical quantity is **the possibility of predicting it with certainty, without disturbing the system**.

In quantum mechanics, in the case of two physical quantities described by non-commuting operators, the knowledge of one precludes the knowledge of the other.

Then, either

1. The description of reality given by the wavefunction in quantum mechanics is not complete;
2. These two quantities cannot have simultaneous reality.

Bell's theorem

Bell's theorem assumes a local hidden variable model, whereby

1. "local" refers to the principle of locality, i.e. an object is influenced directly only by its immediate surroundings. In order for a cause at one point to have an effect at another point, something in the space between those points must mediate the action.
2. "Hidden variable" is a deterministic model to explain the probabilistic nature of quantum theory by introducing additional, possibly inaccessible, variables.

The correlations between the outcomes must obey a specific mathematical constraint, called Bell inequality.

Consider the following setup:

1. Alice and Bob stand in widely separated locations. A pair of particles is prepared. One particle is sent to Alice, the other to Bob.
2. Alice and Bob are free to choose two possible measurements, A_0 and A_1, B_0 and B_1, respectively. Each of these measurements produce ± 1.
3. Suppose that each measurement reveals a property that the particle already possessed. For example, if Alice chooses to measure A_0 and gets $+1$, then the particle she received has $+1$ for a_0.
4. Consider the combination for all possible properties a_0, a_1, b_0, b_1:

$$a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1 = (a_0 + a_1) b_0 + (a_0 - a_1) b_1. \qquad (130)$$

5. Because a_0 and a_1 takes the values ± 1, therefore it is either $a_0 = a_1$ or $a_0 = -a_1$. In the former case, $(a_0 - a_1) b_1 = 0$, while in the latter case, $(a_0 + a_1) b_0 = 0$. One of the terms on the RHS will vanish, and the remaining term will give ± 2.
6. If the experiment repeats many times, the average will be given by

$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2. \qquad (131)$$

Quantum entanglement violates this formulation of Bell inequality, called Clauser-Horne-Shimony-Holt (CHSH) inequality.
Consider the bell state,

$$|\psi\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \tag{132}$$

The first qubit is passed to Alice, while the second to Bob. Alice and Bob measure the following observables,

$$A_0 = \sigma_z, \; A_1 = \sigma_x; \; B_0 = -\frac{\sigma_x + \sigma_z}{\sqrt{2}}, \; B_1 = \frac{\sigma_x - \sigma_z}{\sqrt{2}}.$$

The expectation values are

$$\langle A_0 \otimes B_0 \rangle = \langle A_0 \otimes B_1 \rangle = \langle A_1 \otimes B_0 \rangle = \frac{1}{\sqrt{2}}, \; \langle A_1 \otimes B_1 \rangle = -\frac{1}{\sqrt{2}}.$$

Therefore, the CHSH inequality is violated,

$$\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle = 2\sqrt{2}. \tag{133}$$

Quantum resource theory

In studying algorithms, three questions concern us:

1. What resources are required to perform a given computational task?
2. What computational tasks are possible?
3. What are the limitations (lower bounds) of the number of operations that must be performed by any algorithm?

Quantum resource theory studies quantum information processes under a restricted set of physical operations.

The permissible operations are "free". Since they do not encompass all physical processes, only certain physically realizable states can be prepared. These accessible states are "free".

The "prohibited" operations create "resource" states.

In quantum computing, entanglement is considered as a resource. The "free" operations are local quantum operation and classical communication (LOCC). Entanglement cannot be generated by LOCC.

QMS Journal Club 8th meeting

Motivation

A Turing machine is an idealized computer with a simpler set of basic instructions and an idealized unbounded memory.

We are addressing the second question: What computational tasks are possible?

Hilbert's *Entscheidungsproblem*, or the decision problem, asks whether or not there exists some algorithm which could be used to solve all the problems of mathematics.

The answer to this is no, proved by Church and Turing through Church-Turing thesis.

Turing machines

A Turing machine contains four elements:

1. A program;
2. A finite state control, which acts like a microprocessor;
3. A tape, which acts like a computer memory;
4. A read-write tape-head, which points to the position on the tape.

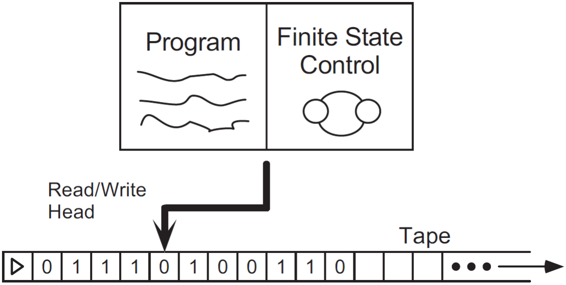


Figure 5: A Turing machine.

The finite state control consists of a finite set of internal states, $q_1, \ldots, q_m$.

There are two special internal states, q_s and q_h, referring to the starting state and halting state respectively.

At the beginning of the computation, Turing machine is in the starting state q_s.

If the computation finishes, the Turing machine ends up in the halting state q_h.

The tape is a one-dimensional object which stretches off to infinity in one direction.

The tape squares contain one symbol drawn from some alphabets 0, 1, b, $\triangleright$. $\triangleright$ marks the left edge of the tape.

The tape-head identifies a single square on the tape as the square that is currently being accessed by the machine.

A program for a Turing machine is a list of program lines of the form $\langle q, x, q', x', s\rangle$.

q is the state from the set of internal states of the Turing machine, while x is the alphabet of symbols on the tape, 0, 1, b, $\triangleright$.

Each cycle, the Turing machine looks through the program lines in order, searching for a line $\langle q, x, \cdot, \cdot, \cdot\rangle$, such that the current internal state is q and the symbol being read is x.

If the machine cannot find such a line, the internal state is changed to q_h, and the machine halts operation.

If such a line is found, then the program line is executed.

Execution of the program is as follows:

1. The internal state is changed to q';
2. The symbol x is overwritten by x';
3. The tape-head moves left, right, or stands still, depending on whether s is -1, $+1$, 0 respectively. The only exception is if the tape-head is at the leftmost square and $s = -1$, then the tape-head stays put.

Let T be a Turing machine defined by the five tuples:

$\langle q_s, \triangleright, q_1, \triangleright, +1\rangle$, $\langle q_1, 0, q_1, b, +1\rangle$, $\langle q_1, 1, q_1, b, +1\rangle$, $\langle q_1, b, q_2, b, -1\rangle$, $\langle q_2, b, q_2, b, -1\rangle$, $\langle q_2, \triangleright, q_3, \triangleright, +1\rangle$, $\langle q_3, b, q_h, 1, 0\rangle$

For each of these initial tapes, determine the final tape when T halts.

1. $\triangleright$, 0, 0, 1, 1, b
2. $\triangleright$, 1, 0, 1, 0, b

What is this Turing machine trying to compute?

Let T be the Turing machine defined by the five-tuples:

$\langle q_0, 0, q_1, 1, +1\rangle$, $\langle q_0, 1, q_1, 0, +1\rangle$, $\langle q_0, b, q_1, 0, +1\rangle$, $\langle q_1, 0, q_2, 1, -1\rangle$, $\langle q_1, 1, q_1, 0, +1\rangle$, $\langle q_1, b, q_2, 0, -1\rangle$

For each of these initial tapes, determine the final tape when T halts.

1. 0, 0, 1, 1, b, b
2. 1, 0, 1, b, b, b
3. b, b, b, b, b, b

What does the Turing machine described by the five-tuples

$\langle q_0, 0, q_0, 0, +1\rangle$, $\langle q_0, 1, q_1, 0, +1\rangle$, $\langle q_0, b, q_2, b, +1\rangle$, $\langle q_1, 0, q_1, 0, +1\rangle$, $\langle q_1, 1, q_0, 1, +1\rangle$, $\langle q_1, b, q_2, b, +1\rangle$

do when given the following tapes as input?

1. 1, 1, b, b, b, b, b, b
2. an arbitrary bit string

Construct a Turing machine with tape symbols 0, 1, b that, when given a bit string as input, replaces the first 0 with a 1 and does not change any of the other symbols on the tape.

Construct a Turing machine that recognizes the set of all bit strings that contain an even number of 1s.

Describe a Turing machine to add two binary numbers x and y modulo 2. The numbers are input on the Turing machine tape in binary, in the form x, followed by a single blank, followed by y. If one number is not as long as the other, then you may assume that it has been padded with leading 0s to make the two numbers the same length.

The initial state of the tape is used to represent the input to the function, and the final state of the tape is used to represent the output of the function.

The Turing machine completely captures the notion of computing a function using an algorithm.

Church-Turing thesis: The class of functions computable by a Turing machine corresponds exactly to the class of functions which we would naturally regards as being computable by an algorithm.

Quantum computer also obey the Church-Turing thesis. The difference is the efficiency with which the computation of the function may be performed. Some functions can be computed much more efficiently on a quantum computer.

There are different types of Turing machines. They are equivalent in the sense that each model is able to simulate the other. A universal Turing machine is a Turing machine that can simulate all other Turing machines and hence any possible computation.

The Halting problem

The Halting problem asks if the Turing machine with number x halt upon input of the number y, i.e. if the algorithm halts or not.

Turing defined the halting function,

$$h(x) = \begin{cases} 0 & \text{if machine number } x \text{ does not halt upon input of } x \\ 1 & \text{if machine number } x \text{ halts upon input of } x \end{cases} \tag{134}$$

If there is an algorithm to solve the halting problem, then there is an algorithm to evaluate $h(x)$. Consider the algorithm

$y = h(x)$
`if` $y = 0$ `then halt`
`else loop forever`
`end if`

This causes a contradiction. Therefore, the initial assumption that there is an algorithm to evaluate $h(x)$ must have been wrong.

QMS Journal Club 9th meeting

Circuit model of computation

An alternative computational model is by using circuit.
A circuit is made up of wires and logic gates.
A logic gate is a function $f : \{0, 1\}^k \to \{0, 1\}^l$ from some fixed number k of input bits to some fixed number l of output bits.
No loops are allowed in the circuit to avoid possible instabilities.
Some important logic gates are:

1. FANOUT gate: Replace a bit with two copies of itself
2. CROSSOVER gate: The value of two bits are interchanged
3. AND gate
4. OR gate
5. NOT gate
6. NAND gate
7. NOR gate

Additionally, one allows the preparation of ancilla bits to allow extra working space during the computation.

A few logic gates can be used to compute any function $f : \{0, 1\}^k \to \{0, 1\}^l$.

The proof is by induction, starting from the simplest case, called the Boolean function $f : \{0, 1\}^n \to \{0, 1\}$.

There are four possible functions:

1. Identity: Wire
2. Bit flip: NOT gate
3. Constant function $f(x) = 0$: AND with ancilla 0 bit
4. Constant function $f(x) = 1$: OR with ancilla 1 bit

The induction follows by increasing the number of bits through cascading the circuits and logic gates from the above.

The set of such logic gates is called the (complete) universal logic set.

Examples of universal logic set are

1. AND, OR and NOT
2. AND and NOT
3. OR and NOT
4. NAND (minimal set)
5. NOR (minimal set)

Question: Show that the NAND gate can be used to simulate AND, XOR, and NOT gates, provided wires, ancilla bits and FANOUT are available.

Not every classical logic gate can be implemented in a straightforward manner in quantum computing. Due to no-cloning theorem, FANOUT cannot be performed. AND and XOR gates are not invertible (quantum operations are unitary).

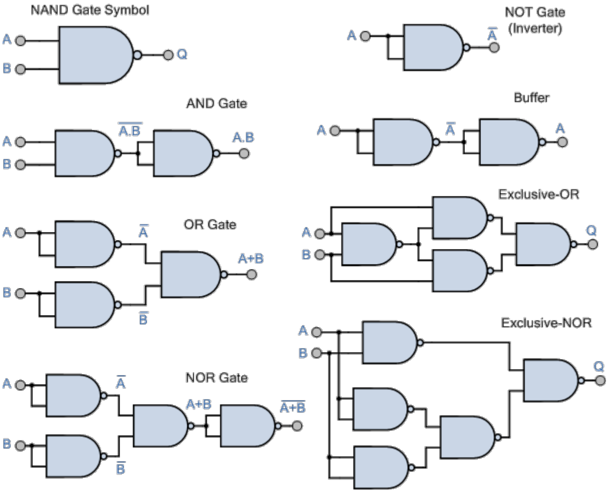


Figure 6: NAND logic gate as the universal gate.

Algorithm-based complexity

Turing machine and the circuit model represent two different models of computation.

On one hand, Turing machine is a uniform model, where the same computational device is used for all possible input lengths.

On the other hand, the circuit model (Boolean circuit) is non-uniform, since inputs of different lengths are processed by different circuits.

The concept of *uniform circuit family* relates both models together by looking at the algorithm-based complexity.

A circuit family consists of a collection of circuits $\{C_n\}$, where C_n is a circuit with n input bits.

A circuit family is uniform if there is some algorithm running on a Turing machine which, upon input of n, generates a description of C_n.

Different models of computation require different computational resources.

One comparison is to look at the asymptotic behavior of a function, without worrying too much about the exact time count.

For example, if a specific algorithm requires $24n + 2\lceil \log n \rceil + 16$ gates to perform, asymptotically the algorithm requires n number of operations in the limit of large problem size.

$O(g(n))$ is used to set the upper bounds $g(n)$ on the behavior of a function $f(n)$, i.e. for sufficiently large n, the function $g(n)$ is an upper bound on $f(n)$.

$O(g(n))$ is useful for studying the worst-case behavior of specific algorithms.

$\Omega(g(n))$ examines the lower bounds $g(n)$ on the behavior of a function $f(n)$, i.e. for sufficiently large n, $g(n)$ is a lower bound on $f(n)$.

$\Theta(g(n))$ indicates that $f(n)$ behaves the same as $g(n)$ asymptotically. $f(n)$ is $\Theta(g(n))$ if it is both $O(g(n))$ and $\Omega(g(n))$.

Example 1: $f(n) = 2n$ is $O(n^2)$, since $2n \leq 2n^2$ for all positive n.

Example 2: $f(n) = 2^n$ is $\Omega(n^3)$, since $n^3 \leq 2^n$ for sufficiently large n.

Example 3: $f(n) = 7n^2 + \sqrt{n}\log n$ is $\Theta(n^2)$, since $7n^2 \leq f(x) \leq 8n^2$ for sufficiently large n.

Exercise 1: Show that n^k is $O(n^{\log n})$ for any k, but that $n^{\log n}$ is never $O(n^k)$.

Exercise 2: Show that c^n is $\Omega(n^{\log n})$ for any $c > 1$, but that $n^{\log n}$ is never $\Omega(C^n)$.

A simple compare-and-swap algorithm for solving the sorting problem is as follows.

```
for j = 1 to n-1
    for k = j+1 to n
       compare-and-swap(j,k)
    end k
end j
```

This algorithm uses

$$(n-1)+(n-2)+\ldots+1=\frac{n(n-1)}{2}$$

operations, hence it belongs to Θn^2.

Computational complexity

Computational complexity studies the lower bounds on the resources required by the best possible algorithm for solving a problem, even if the algorithm is not explicitly known.

Ideally, the most efficient algorithm would match with the lower bound proved by computational complexity, but this is often not the case.

A coarse classification of computational problems is to ask whether or not a problem can be solved using resources bounded by a polynomial in n (**P**), or exponential (**NP**).

A problem is regarded as *easy, tractable, or feasible* if an algorithm for solving the problem using polynomial resources exists. Otherwise, it is *hard, intractable, or infeasible*.

P and **NP** are theoretical constructs. In reality, an algorithm using more operations is probably more useful than an optimized and efficient algorithm.

Strong Church-Turing thesis: Any model of computation can be simulated on a probabilistic Turing machine with at most a polynomial increase in the number of elementary operations required.

A probabilistic Turing machine is a non-deterministic Turing machine that chooses between the available transitions at each point according to some probability distribution. It was observed to be the strongest reasonable model of computation.

In other words, if a problem has no polynomial resource solution on a probabilistic Turing machine, then it has no efficient solution on any computing device.

QMS Journal Club 10th meeting

Complexity classes

Many computational problems can be framed as decision problems, primarily because the problems take their simplest form for analysis, but generalizable to other complex scenarios.

A language L over the alphabet $\sum$ is a subset of the set $\sum^*$ of all (finite) strings of symbols from $\sum$.

Example: To convert the primarity decision problem into formal language, the binary alphabet is $\sum = \{0, 1\}$. $\sum^*$ can be interpreted as non-negative integers. L is defined to consist of all prime numbers in binary strings.

Turing machine can be "tuned" to have two halting states, q_Y and q_N, to represent 'yes' and 'no' respectively.

A language L is decided by a Turing machine if the machine is able to decide whether an input x on its tape is a member of the language of L or not, eventually halting in the state q_Y if $x \in L$, and q_N if $x \notin L$.

A problem is in **TIME**$(f(x))$ if there exists a Turing machine which decides whether a candidate x is in the language in time $O(f(n))$, where n is the length of x.

Previously, we mentioned a coarse classification of computational problems by looking at the computational resources, either bounded by a polynomial in n (**P**) or exponential (**NP**).

An **NP** problem is that the 'yes' instances of a problem can be easily verified with the aid of an appropriate *witness*, while there is no need to produce a witness to $x \notin L$.

The asymmetry of **NP** prompted a new class of languages, **coNP** which have witnesses to 'no' instances.

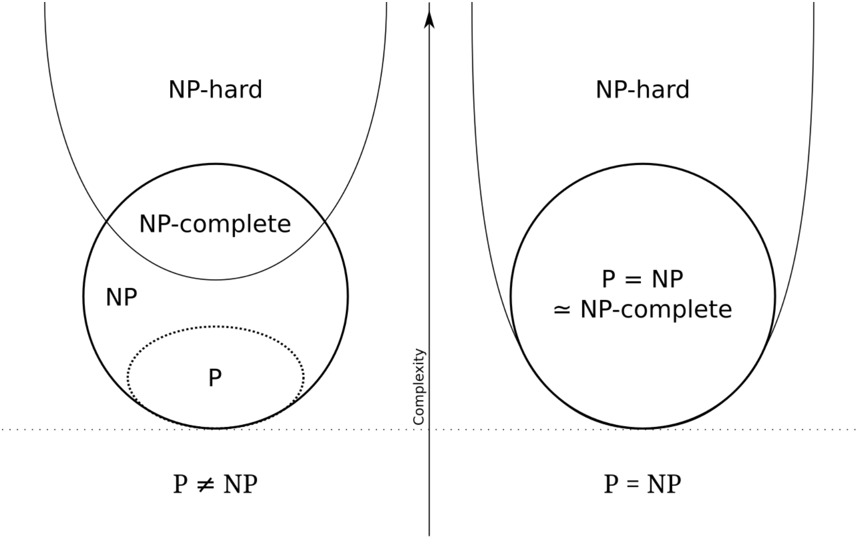


Figure 7: **P** $\neq$ **NP** problem.

A cycle is a sequence of vertices, $v_1, \ldots, v_m$ such that each pair of vertices (v_j, v_{j+1}) defines an edge.

A simple cycle is a cycle in which none of the vertices is repeated, except for the first and last vertices.

A Hamiltonian cycle is a simple cycle which visits every vertex in the graph.

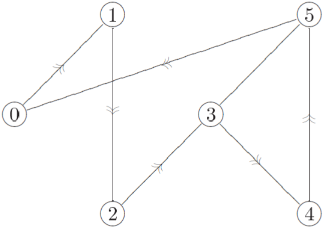


Figure 8: A Hamiltonian cycle.

The Hamiltonian cycle problem determines whether a given graph contains a Hamiltonian cycle or not.

It is a problem in the class of **NP**-complete problems. Any other **NP** problems can be reduced to **NP**-complete problems in polynomial time, and that the solution is verifiable in polynomial time.

If any **NP**-complete problem has a polynomial time solution, then $\mathbf{P} = \mathbf{NP}$.

The Hamiltonian cycle problem can be reduced to the traveling salesman problem, which asks that given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?

Suppose we have a graph of n vertices. By defining $d_{ij} = 1$ if vertices i and j are connected, and $d_{ij} = 2$ if vertices i and j are not connected, we convert the traveling salesman problem into the Hamiltonian cycle problem by finding the distance $d = n + 1$. If a tour is of distance $d = n$, then it is a Hamiltonian cycle. Conversely is true.

A language B is said to be reducible to another language A if there exists a Turing machine operating in polynomial time such that given input x and output $R(x)$, $x \in B$ if and only if $R(x) \in A$.

It means that if we have an algorithm for deciding A, then with a little extra overhead we can decide the language B.

There is a language L in the complexity class which is the “most difficult” to decide, in the sense that every other language in the complexity class can be reduced to L. This is called a *complete* problem.

An Euler cycle is an ordering of the edges of a graph G so that every edge in the graph is visited exactly once.

The Euler cycle problem determines whether a given graph contains a Euler cycle or not.

The Euler cycle problem is exactly the same problem as the Hamiltonian cycle problem, the only difference being that vertices are changed to edges.

Theorem: A connected graph contains an Euler cycle if and only if every vertex has an even number of edges incident upon it.

The theorem above provides a method to efficiently solve the Euler cycle problem, and thus the Euler cycle problem is in **P**.

Assuming $\mathbf{P} \neq \mathbf{NP}$, another class of problems called **NPI**, which is neither solvable with polynomial resources nor **NP**-complete is proposed.

This is of interest to quantum information physicists because many suspect that quantum computers will not be able to efficiently solve all problems in **NP**.

Two candidate problems in **NPI** are the factoring and graph isomorphism problems.

Graph isomorphism problem: Suppose G and G' are two undirected graphs over the vertices $V = \{v_1, \ldots, v_n\}$. Does there exist a one-to-one function $\phi : V \to V$ such that the edge (v_i, v_j) is contained in G if and only if $(\phi(v_i), \phi(v_j))$ is contained in G'?

Sometimes, instead of finding exact solutions to an **NP**-complete problem, we can try to find approximate solutions.

However, the reduction of approximate solution to an **NP**-complete problem may not preserve the notion of "good approximation" to another **NP** problem.

The space-bounded complexity class, **PSPACE**, studies decision problems which may be solved on a Turing machine using a polynomial number of working bits, with no limitation on the amount of time that may be used by the machine.

The complexity class **EXP** contains all decision problems which may be decided by a Turing machine running in exponential time.

The complexity class **L** contains all decision problems which may be decided by a Turing machine running in logarithmic space.

Does allowing more time or space give greater computational power?

Currently, we know that

$$\mathbf{L} \subseteq \mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{PSPACE} \subseteq \mathbf{EXP} \tag{135}$$

but there is no proof that the inclusions are strict.

In distributed computing, the communication complexity studies the cost of communication between the units in solving a problem.

Energy and computation

Energy consumption in computation is related to the reversibility of the computation.

Landauer's principle (first form): Suppose a computer erases a single bit of information. The amount of energy dissipated into the environment is at least $k_B T \ln 2$.

Landauer's principle (second form): Suppose a computer erases a single bit of information. The entropy of the environment increases by at least $k_B \ln 2$.

Landauer's principle provides a lower bound on the amount of energy that must be dissipated to erase information.

An implication from Moore's law is that if computer power keeps increasing, then the amount of energy dissipated must also increase, unless the energy dissipated per operation drops at least as fast as the rate of increase in computing power.

If all computations could be done reversibly, then Landauer's principle would imply no lower bound on the amount of energy dissipated by the computer.

Remember that AND and OR gates are irreversible gates. The irreversibility may be viewed as a consequence of some ancilla bits being "hidden" and erased during the execution of the gates.

In the process of building reversible gates with ancilla bits, we use a technique called "uncomputation" to undo all the computations done on the ancilla bits so that they can be reused. Since all operations are reversible, they do not cost energy (but some extra operational time).

The complexity classes of **P** and **NP** do not change whether a reversible or irreversible model of computation is used.

Toffoli gate as the universal reversible gate

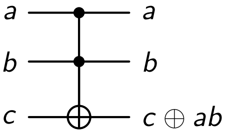


Figure 9: Toffoli gate (or CCNOT gate).

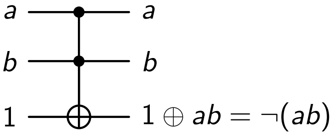


Figure 10: NAND gate.

Since NAND gate is a minimal universal gate set, Toffoli gate can be used to build all logic gates in the reversible computation model.

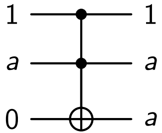


Figure 11: FANOUT gate.

Because the computer's memory is finite, error correction codes must eventually be erased so that it can be recycled. This erasure carries some energy costs.